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Bachelor's Thesis in Computer Science

Data-driven camera calibration for light field camera systems

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Abstract

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1. Introduction

1.1. Motivation

1.2. Challenges and Problem Statement

1.3. Research Questions

The following research questions guide this thesis:

- **RQ1:** Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?
- **RQ2:** How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?
- **RQ3:** Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?

1.4. Structure of the Thesis

2. Theoretical Background

2.1. Pinhole Camera Model

Cameras are the primary devices used to capture visual information from the real world into an image. To process and analyze this information computationally, it is necessary to describe the imaging process using a camera model [10].

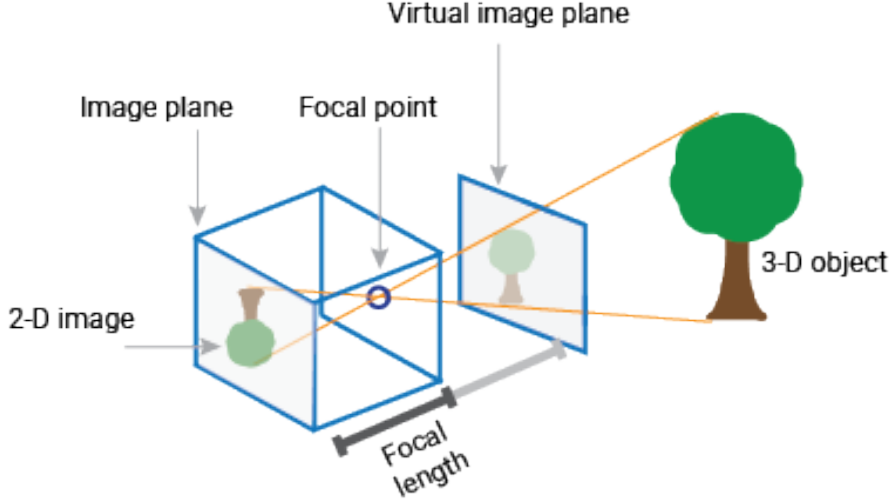


Figure 1: Illustration of the pinhole camera model. Source: [12].

The pinhole camera model describes a simple imaging system in which light rays from a 3D scene pass through a small aperture and are projected through the focal point onto a 2D image plane [10]. The aperture limits the incoming light such that each point in the scene is mapped to a single point on the image plane. In this model, the projected image is inverted due to the geometry of the projection, as shown in Figure 1.

To define the mapping from 3D world coordinates to 2D image coordinates, the model is mathematically described using two types of parameters [12].

- **Intrinsic Parameters:**

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

where (f_x, f_y) are the focal lengths in pixel units along the x and y axes, s is the skew coefficient between the axes, and (c_x, c_y) is the principal point (optical center) in pixel coordinates. K is also known as the camera matrix.

- **Extrinsic Parameters:**

$$[R \mid t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

where R is a 3×3 rotation matrix representing the orientation of the camera with respect to the world coordinate system, and t is a 3×1 translation vector representing the position of the camera center in world coordinates.

These parameters form the basis of camera calibration, which aims to estimate them accurately. However in practice, cameras use lenses to focus light onto the image sensor, which introduces distortions not accounted for in the idealized pinhole model.

2.2. Lens Distortion

The camera matrix alone is not sufficient to model real imaging systems, as the ideal pinhole camera model assumes a lens-free projection. In practice, however, camera lenses introduce various optical distortions that must be taken into account to accurately describe the image formation process [12].

Lens distortion is defined as an optical aberration that causes straight lines in the real world to appear curved in the captured image [4]. Since distortion alters the geometric relationship between 3D world points and their 2D image projections, it has a direct impact on the accuracy of camera calibration. If lens distortion is not properly modeled, the estimated intrinsic and extrinsic parameters can become biased, leading to unreliable calibration results [15].

In most camera calibration models, lens distortion is represented using a small set of parametric functions. The two most common types of distortion considered in practice are radial distortion and tangential distortion.

2.2.1. Radial Distortion

Radial distortion is caused by the spherical shape of camera lenses, which leads to different refraction of light from the image center to the outer regions. This causes straight lines in the image to appear curved, resulting in barrel or pincushion distortion [3].

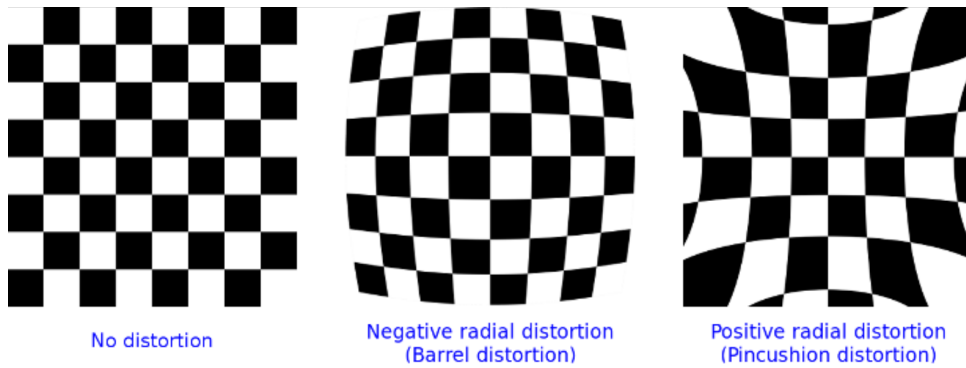


Figure 2: Comparison of different types of radial distortion. Source: [14].

The formal definition of radial distortion is given as follows [12]:

$$x_{distorted} = x \left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 \right) \quad (1)$$

$$y_{distorted} = y \left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6 \right) \quad (2)$$

where x and y denote the undistorted normalized image coordinates in the camera coordinate system. The radius r represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \quad (3)$$

The coefficients k_1 , k_2 , and k_3 in (1) and (2) control the strength of radial lens distortion. In practice, the first coefficient k_1 has the dominant influence on the distortion type: negative values of k_1 typically result in barrel distortion, while positive values lead to pincushion distortion.

2.2.2. Tangential Distortion

Tangential distortion occurs when the image sensor and the camera lens are not perfectly aligned or parallel to each other [12]. As a result, image points are displaced tangentially with respect to the optical axis. Figure 3 illustrates the geometric cause of tangential distortion and its effect on the captured image.

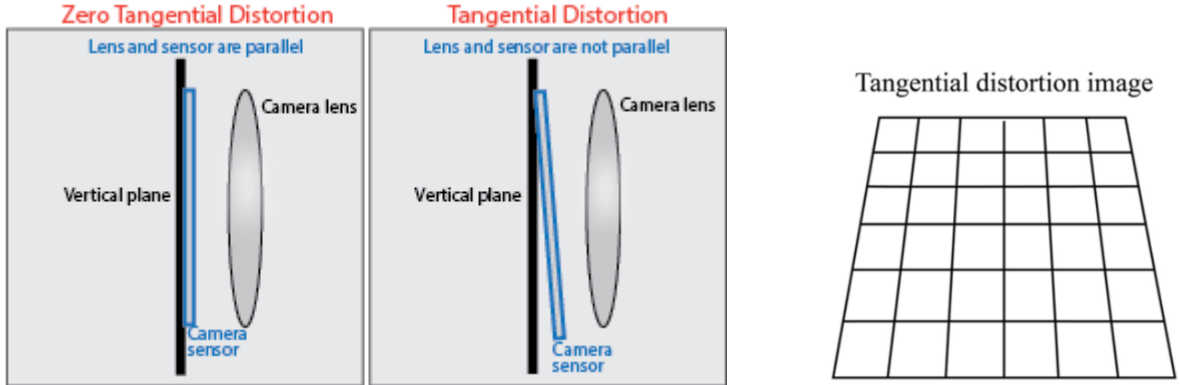


Figure 3: Visualization of tangential distortion. Source: [12], [11].

The formal definition of tangential distortion is given by the following equations [12]:

$$x_{distorted} = x + \left(2p_1 xy + p_2 (r^2 + 2x^2) \right) \quad (4)$$

$$y_{distorted} = y + \left(p_1 (r^2 + 2y^2) + 2p_2 xy \right) \quad (5)$$

Here, x and y denote the undistorted normalized image coordinates in the camera coordinate system. The radius r represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \quad (6)$$

The coefficients p_1 and p_2 describe how strong the tangential lens distortion is. This type of distortion is caused by mechanical misalignment of the lens elements relative to the image sensor, resulting in a displacement of image points. Although tangential distortion is typically less pronounced than radial distortion, it can still negatively affect the camera calibration accuracy if not properly modeled.

2.3. Camera Calibration

The goal of camera calibration, also known as camera resectioning, is to estimate the parameters of a camera model. These parameters include the intrinsic parameters, the extrinsic parameters, and the lens distortion coefficients [12]. Knowing these parameters makes it possible to correct distortions in the recorded images and to perform metric measurements in the scene, for example to estimate object sizes or distances from image data.

Camera calibration is required for many tasks in 3D computer vision and photometry. Applications such as depth estimation, 3D reconstruction, or pose estimation rely on a precise mapping from 3D world coordinates to 2D image coordinates [16]. If the calibration is inaccurate, errors can propagate through the processing pipeline and reduce the quality of the final results.

In practice, cameras are often calibrated using targets with known geometric properties. Planar targets, such as checkerboard patterns, are widely used because they are easy to handle and lead to reliable results [18]. Several images of the target are captured from different viewpoints and orientations. Based on these images, the camera parameters can be estimated with relatively small calibration errors. For this reason, target-based calibration methods are commonly used in many computer vision applications.

2.3.1. Target-Based Camera Calibration

There are various approaches for target-based camera calibration. One of the most common methods uses a checkerboard pattern, as mentioned before. The checkerboard consists of a regular grid of black and white squares with known dimensions, as shown in Figure 4.

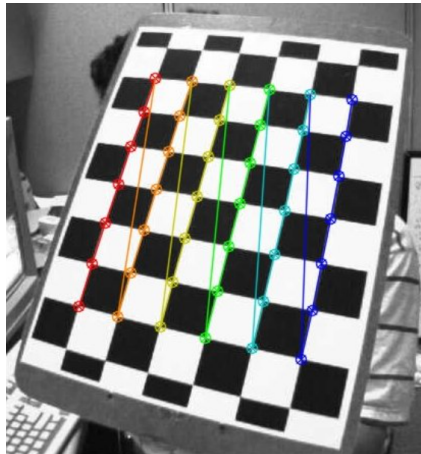


Figure 4: Example of a checkerboard and detected corner points. Source: [13].

During the calibration process, control points are extracted from the images. In the case of a checkerboard, these control points correspond to the corner points of the squares [16]. The positions of these corners are known in the coordinate system of the calibration target. By capturing multiple images of the checkerboard from different angles and distances, a set of 2D image points can be obtained that correspond to the known 3D points of the target. Based on these point correspondences, an optimization problem can be formulated to estimate the intrinsic and extrinsic camera parameters as well as the lens distortion coefficients [18].

This calibration approach is implemented in widely used computer vision libraries such as OpenCV [13]. It can be applied to single-camera setups and multi-camera systems. Besides checkerboards, other types of calibration targets exist, for example circular dot patterns [16]. In some cases, these patterns can achieve higher accuracy, especially when the image quality is low. However, they also require physical targets and a manual calibration setup.

The main limitations of target-based calibration methods are the required manual placement of the target and the need to capture images with different poses. In addition, the calibration accuracy depends on factors such as the quality of the corner detection, the number of images used, and the diversity of viewpoints during data acquisition [18]. Despite these limitations, target-based calibration remains a standard method in many computer vision applications due to its simplicity and reliable performance.

2.3.2. Calibration Parameters and Error Metrics

To evaluate the quality of a camera calibration, different error metrics are used. These metrics describe how well the estimated camera parameters fit the observed image data. In practice, this is done by comparing detected image points with points that are projected using the calibrated camera model. The most common metric is the reprojection error [12].

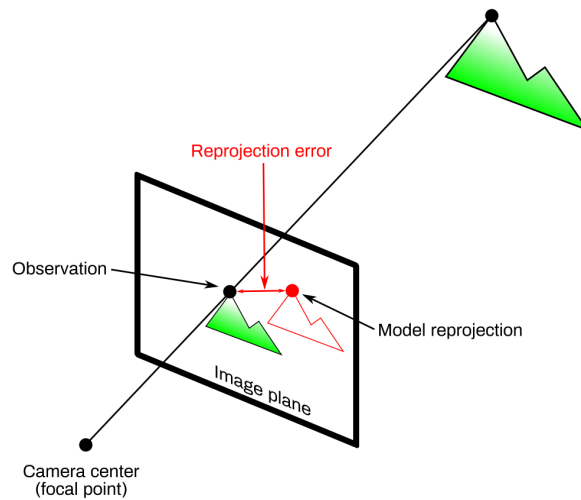


Figure 5: Visualization of reprojection error. Source: [1].

After a calibration, known 3D points are projected back onto the image plane. The reprojection error measures the distance between the observed image points and the

corresponding projected points. It is usually computed as the average Euclidean distance and is given in pixel units [1]. It can be defined as

$$e = \frac{1}{N} \sum_{i=0}^{N-1} \|\mathbf{p}_i - \mathbf{q}_i\|_2. \quad (7)$$

Here, $\mathbf{p}_i$ represents the observed feature points in the image, while $\mathbf{q}_i$ denotes the projected points obtained from the calibrated camera model. In general, a smaller reprojection error indicates a better calibration. What is considered a good value depends on the application, but for accurate calibrations the reprojection error is often between 0.1 and 0.5 pixels [8].

Another commonly used measure is the root mean square (RMS) error [7]. It summarizes the reprojection error over all calibration points and is computed as:

$$\epsilon_{\text{res}} = \left(\frac{1}{2n} \sum_{i=1}^n \|\mathbf{p}'_i - \mathbf{q}'_i\|_2^2 \right)^{1/2} \quad (8)$$

The RMS error provides a single value that describes the overall calibration accuracy and is often used to compare different calibration results.

2.3.3. Bundle Adjustment

Bundle adjustment is a method for improving a 3D reconstruction by refining both the positions of 3D points and the camera parameters at the same time [17]. The name comes from the “bundles” of light rays from the 3D points to the cameras, which are adjusted together to make the reconstruction more accurate. The parameters that are optimized together usually include the 3D coordinates of the points, the extrinsic parameters, and sometimes the intrinsic parameters [2].

This joint optimization improves calibration accuracy because it ensures that all cameras and points are consistent with each other. As a result, projection errors are reduced and small misalignments are corrected that would remain if cameras or points were adjusted separately [17]. This idea is illustrated in Figure 6.

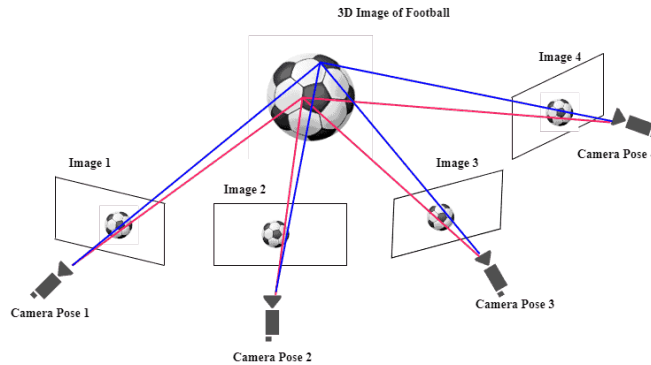


Figure 6: Visual representation of bundle adjustment. Source: [9].

Mathematically, bundle adjustment is formulated as an optimization problem that minimizes the reprojection error. This optimization is usually solved using nonlinear least squares methods and takes advantage of the fact that the problem structure is largely sparse [2].

2.4. Light Field Camera Systems

A standard camera captures a 2D projection of the 3D world. Light field cameras, on the other hand, have the ability to capture more information. They record not only the intensity of light rays but also their direction [6]. This property allows the creation of accurate depth maps and refocus the image after it has been taken [5]. Therefore light field cameras are growing in importance for applications in computer vision, robotics, augmented reality, and other applications.

The four-dimensional nature of light fields means that their processing currently needs relatively high computational resources [6]. However, the evolution of hardware and algorithms is making light field cameras more practical for real-world applications.

There are mainly two different types of light field camera systems: multi-camera arrays and micro-lens-array-based cameras. Each type has its own advantages and disadvantages, which will be discussed in the following sections.

2.4.1. Multi-Camera Light Field Systems

How is a multi-camera light field system structured?

How do multiple cameras sample the light field?

What are the main advantages of camera arrays?

What are the main challenges of multi-camera setups?

2.4.2. Micro-Lens-Array-Based Light Field Cameras

How does a micro-lens array capture a light field?

How is spatial and angular information encoded?

What are the main advantages of MLA-based cameras?

What are the main limitations compared to camera arrays?

2.5. State of the Art in Light Field Camera Calibration

3. Material and Method

3.1. Material

3.1.1. Synthetic Data Generation with Blender

3.1.2. Image Sets and Ground Truth Definition

3.1.3. Runtime Environment and Toolchain

3.2. Method

3.2.1. Use Case Description

3.2.2. Requirements Analysis

3.2.3. UML Class Diagram

3.2.4. Health Monitoring and Decision Metrics

3.2.5. State Diagram

4. Results and Evaluation

4.1. Results of Initial Calibration

4.1.1. Intrinsic Accuracy

4.1.2. Extrinsic Accuracy

4.1.3. Reprojection and RMS Error

4.2. Results of LiFCal-Based Recalibration

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4.2.2. Health Score Evolution

5. Discussion of the Results

5.1. Interpretation of Initial Calibration Results

5.2. Limitations of LiFCal in the Given Setup

5.3. Comparison between Initial Calibration and LiFCal

5.4. Comparison to state of the Art

5.5. Potential Extensions and Improvements

6. Conclusion and Outlook

6.1. Conclusion

6.2. Outlook

A. Appendix

Acknowledgment

References

- [1] CamCalib. The reprojection error. <https://www.camcalib.io/post/what-is-the-reprojection-error>, 2022. CamCalib blog, accessed 2026-01-21.
- [2] Daniel Cremers. Bundle adjustment & nonlinear optimization. https://cvg.cit.tum.de/_media/teaching/ss2019/mvg2019/material/multiviewgeometry7.pdf, 2019. Vorlesungsunterlage, Technische Universität München.
- [3] Elsevier. Radial distortion. <https://www.sciencedirect.com/topics/computer-science/radial-distortion>, 2022. Definition from ScienceDirect Topics: Computer Science.
- [4] Elsevier. Lens distortion. <https://www.sciencedirect.com/topics/engineering/lens-distortion>, 2024.
- [5] Aymeric Fleith, Doaa Ahmed, Daniel Cremers, and Niclas Zeller. Lifcal: Online light field camera calibration via bundle adjustment. <https://arxiv.org/abs/2408.11682>, 2024.
- [6] German Aerospace Center (DLR) – Institute of Robotics and Mechatronics. Light field cameras. DLR Research Expertise: 3D Perception, 2026. URL <https://www.dlr.de/en/rm/research/expertise/3d-perception/light-field-cameras>. Accessed January 24, 2026.
- [7] Richard Hartley and Andrew Zisserman. *Multiple View Geometry in Computer Vision*. Cambridge University Press, Cambridge, UK, 2nd edition, 2003. ISBN 0521540518. URL http://www.r-5.org/files/books/computers/algo-list/image-processing/vision/Richard_Hartley_Andrew_Zisserman-Multiple_View_Geometry_in_Computer_Vision-EN.pdf.
- [8] Jiachun Huang, Shaoli Liu, Jianhua Liu, and Zehua Jian. Camera calibration results: Reprojection error and relative positions. *Scientific Reports*, 2016. URL https://www.researchgate.net/figure/Camera-calibration-results-a-Reprojection-error-and-b-relative-positions-of-the_fig9_301715405. In Fig. 5.
- [9] Dhiraj Kumar. What is bundle adjustment? Baeldung on Computer Science, 2024. URL <https://www.baeldung.com/cs/computer-vision-bundle-adjustment>. Last updated March 18, 2024.
- [10] Fei-Fei Li et al. Camera models. https://web.stanford.edu/class/cs231a/course_notes/01-camera-models.pdf, 2020.
- [11] Shih-Hao Lin, En-Tze Chen, Jia-Jun Xiao, and Ching-Yuan Chang. Stereo digital image correlation (3d-dic) for non-contact measurement and refinement of delta robot arm displacement and jerk. *The International Journal of Advanced Manufacturing Technology*, 133(5-6):2809–2822, July 2024. doi: 10.1007/s00170-024-13957-2. URL <https://doi.org/10.1007/s00170-024-13957-2>.
- [12] MathWorks. Camera calibration. <https://de.mathworks.com/help/vision/ug/camera-calibration.html>, 2024.

- [13] OpenCV Contributors. Camera calibration with opencv. https://docs.opencv.org/4.x/dc/dbb/tutorial_py_calibration.html, 2024.
- [14] OpenCV Developers. Camera calibration and 3d reconstruction — opencv documentation. OpenCV 4.x Documentation, calib3d module, 2025. URL https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html.
- [15] Carlos Ricolfe-Viala and Antonio-José Sánchez-Salmerón. Lens distortion models evaluation. *Applied Optics*, 49(30):5914–5928, 2010. doi: 10.1364/AO.49.005914. URL https://www.researchgate.net/publication/47510646_Lens_distortion_models_evaluation.
- [16] Chaehyeon Song, Dongjae Lee, Jongwoo Lim, and Ayoung Kim. Camera calibration via circular patterns: A comprehensive framework with measurement uncertainty and unbiased projection model. *arXiv preprint arXiv:2506.16842*, 2025. URL <https://arxiv.org/abs/2506.16842>.
- [17] Bill Triggs, Philip F. McLauchlan, Richard I. Hartley, and Andrew W. Fitzgibbon. Bundle adjustment – a modern synthesis. In Bill Triggs, Andrew Zisserman, and Richard Szeliski, editors, *Vision Algorithms: Theory and Practice*, volume 1883 of *Lecture Notes in Computer Science*, pages 298–372, London, UK, 2000. Springer-Verlag. doi: 10.1007/3-540-44480-7_21. URL <https://www.cs.jhu.edu/~misha/ReadingSeminar/Papers/Triggs00.pdf>.
- [18] Z. Zhang. A flexible new technique for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(11):1330–1334, 2000. doi: 10.1109/34.888718. URL <https://ieeexplore.ieee.org/document/888718>.